Topic 6: Harmonic model

From sinusoidal tracks to harmonic structure and fundamental-frequency estimation

Digital Signal Processing for Sound and Music · Topic 6

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Lecture roadmap

1 **Harmonic model**
sinusoids constrained by F0

2 **Partials & harmonics**
what changes from the sinusoidal model

3 **Harmonic detection**
use F0 to select peaks

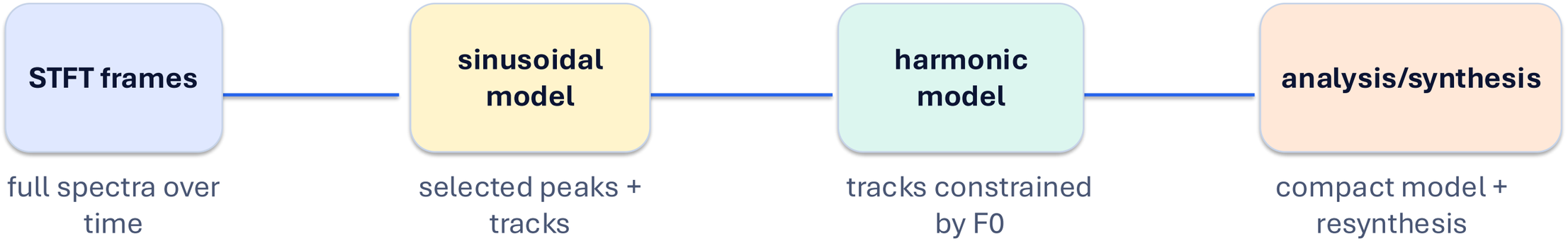
4 **F0 detection**
time-domain and spectral approaches

5 **Examples**
f0-detection and harmonic-model notebooks

6 **Exercise 6**
implement, tune, inspect, listen

From sinusoidal to harmonic representation

The harmonic model adds one important constraint: sinusoidal frequencies follow a common F0



Use harmonic modeling only when the sound source is periodic or pseudo-periodic enough for the harmonic constraint to make sense.

Harmonic model

The frequencies of the sinusoidal components are constrained to be multiples of one fundamental frequency

Time-domain model

$$y_h[n] = \Sigma_r A_r[n] \cos(2\pi\, r\, f_0[n]\, n + \varphi_r[n])$$

r: harmonic number
R: number of harmonic components
$A_r[n]$: instantaneous amplitude
$f_0[n]$: fundamental frequency
$\phi_r[n]$: instantaneous phase

Difference with the sinusoidal model: $f_r[n] = r \cdot f_0[n]$

When is it valid?

- single-source or source-separated signals
- periodic or pseudo-periodic oscillation
- musical instruments or voice with stable pitch
- not a good description of strongly inharmonic components
- less appropriate for dense polyphonic mixtures

The model is more restrictive, but more interpretable.

Harmonic spectrum as shifted window spectra

Each harmonic contributes a window-shaped spectral lobe at a multiple of the fundamental frequency

Frequency-domain expression

$$Y_h[k] = \Sigma_r \, A_r \, W[k - r \, f_0]$$

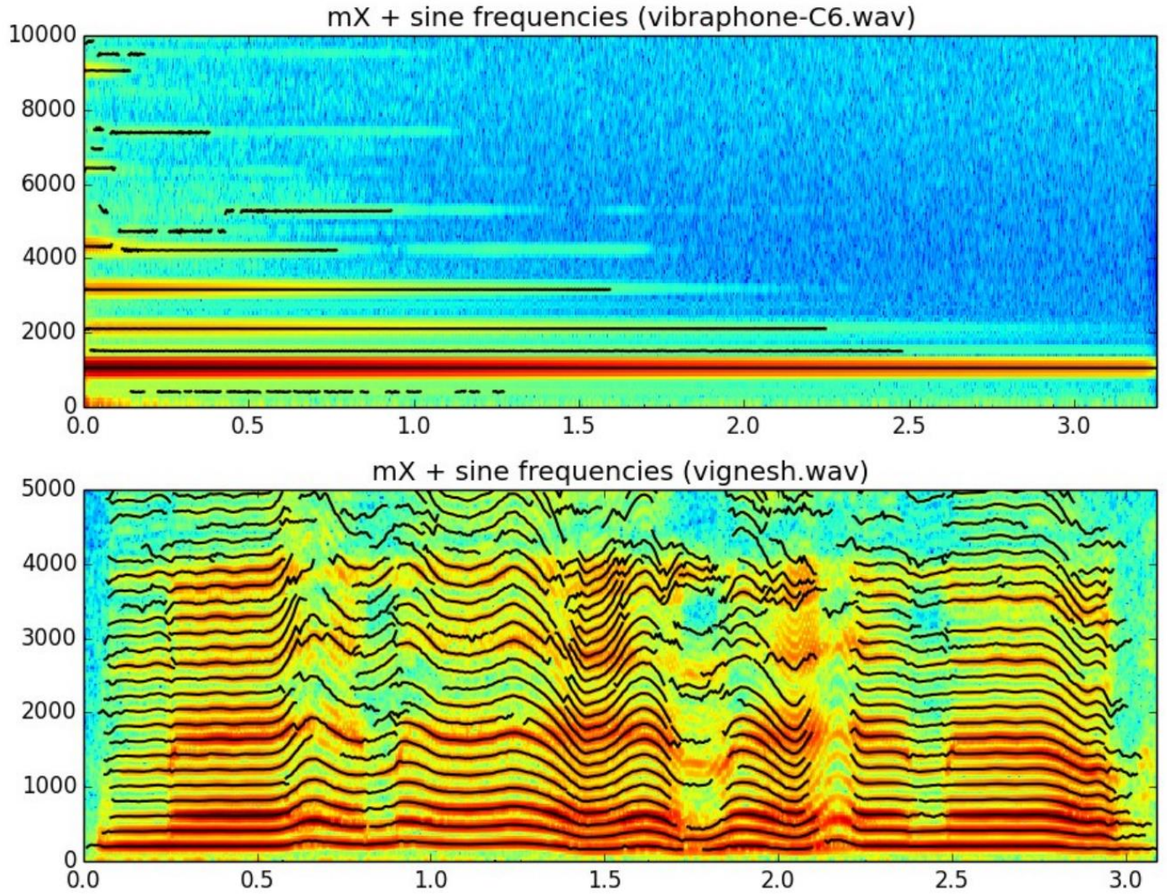
W: spectrum of the analysis/synthesis window
r: harmonic number
f_0: normalized fundamental frequency
A_r: harmonic amplitude
k: DFT-bin index

Interpretation

- a harmonic sound is not just “many peaks”
- peaks should align with an integer harmonic series
- the F0 links all harmonics into one source
- harmonic detection becomes easier after F0 detection
- resynthesis can use the harmonic parameters only

Inharmonic vs harmonic sinusoidal tracks

The same sinusoidal-tracking machinery behaves differently depending on the sound source



What to notice

- vibraphone tracks are not locked to a harmonic series
- voice tracks show many parallel harmonic ridges
- harmonic modeling is appropriate for the second case
- the model choice depends on the source, not just the algorithm

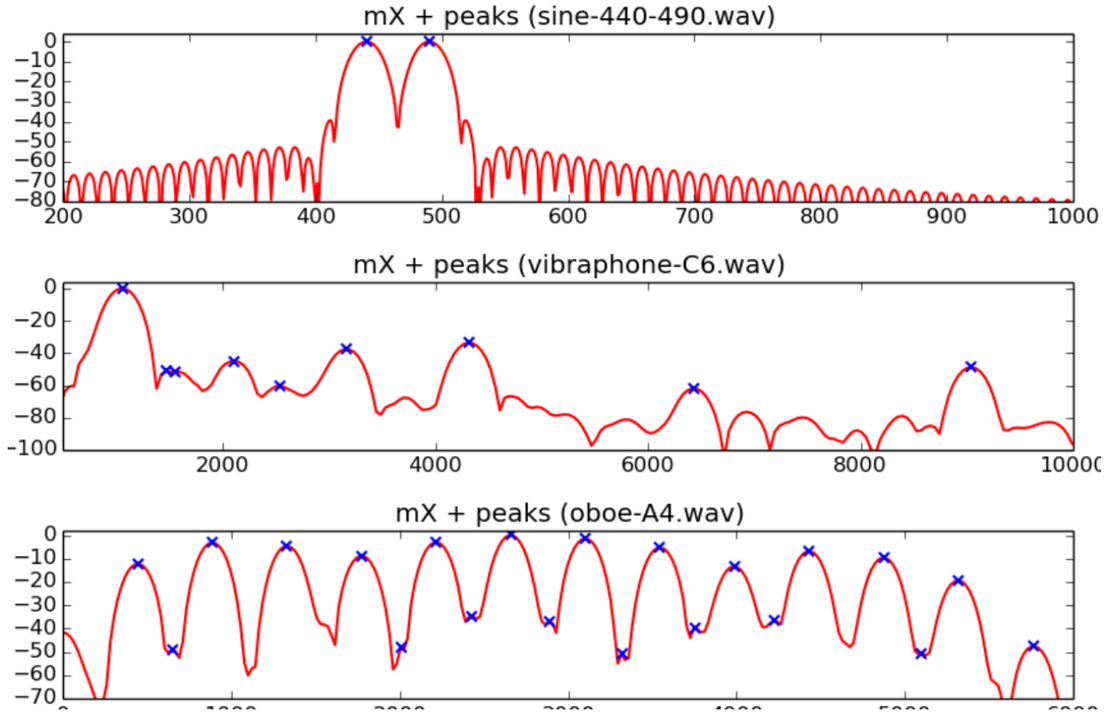
Sinusoids, partials, and harmonics

These terms are related but not interchangeable

Sinusoid a mathematical function with one frequency, amplitude, and phase

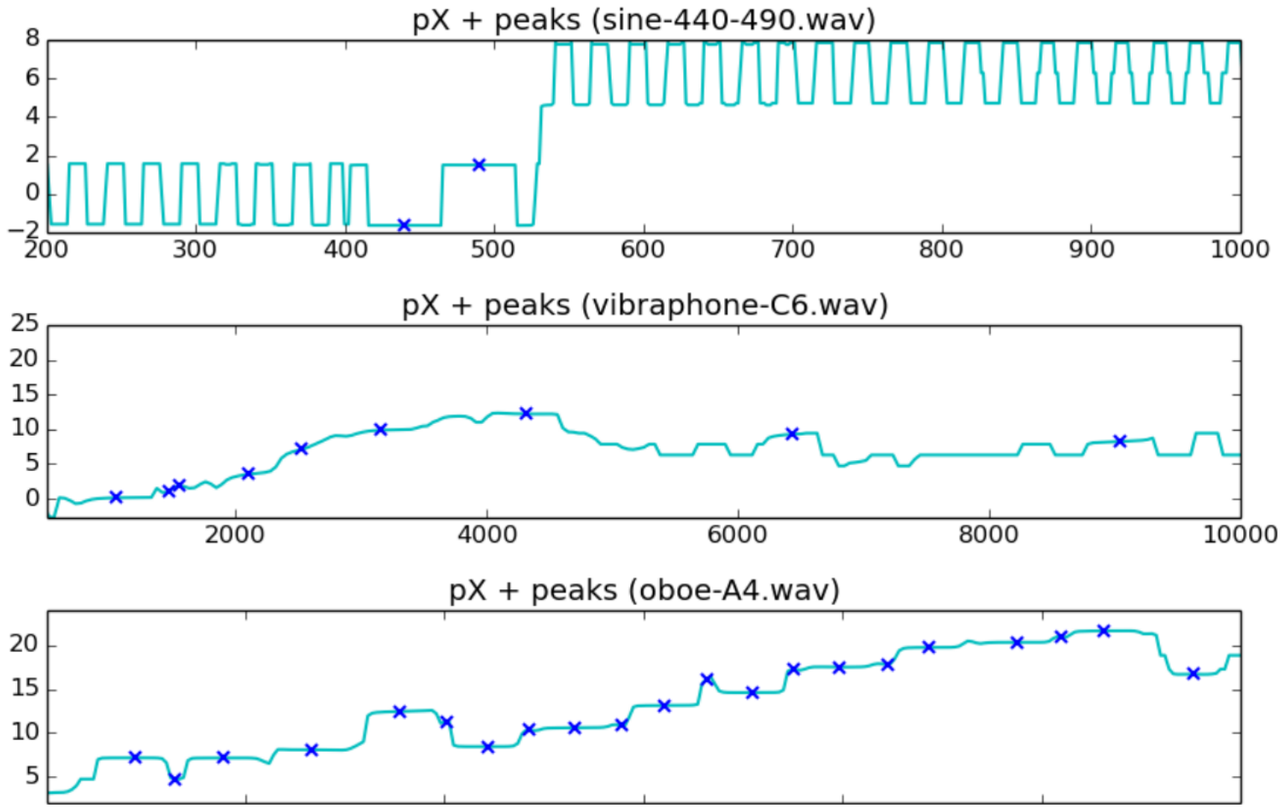
Partial a stable signal component that can be modeled by a slowly-varying sinusoid

Harmonic a partial whose frequency is close to an integer multiple of F0



Phase spectra also reveal differences between components

Magnitude shows where peaks are; phase helps check whether the interpretation is consistent

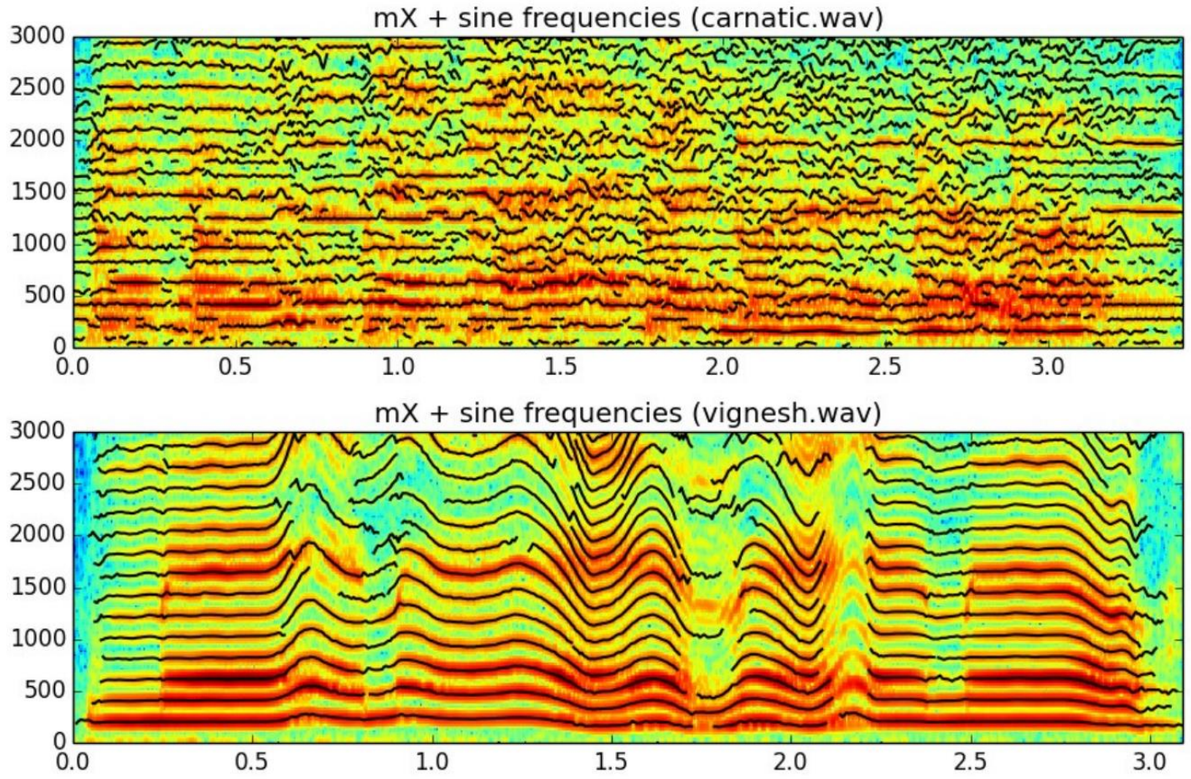


Reading strategy

- start from magnitude to find candidates
- use phase mainly near significant peaks
- compare behavior across related peaks
- do not assume every stable peak is harmonic

Polyphonic and monophonic signals

The harmonic model works most directly on one harmonic source, not on a mixture of sources



Implication

- monophonic: one source to model
- polyphonic: several overlapping sources
- harmonic series may overlap or mask each other
- F0 detection becomes much harder in mixtures

Harmonic detection after F0 detection

A peak becomes a harmonic candidate only if its frequency is close to an integer multiple of F0

Detection rule

peak p is harmonic h if $| f_p - h f_0 | <$ threshold
for $h = 1, 2, \ldots, H$

f_p: detected peak frequency
f_0: fundamental frequency
h: harmonic number
threshold: tolerance in Hz or cents

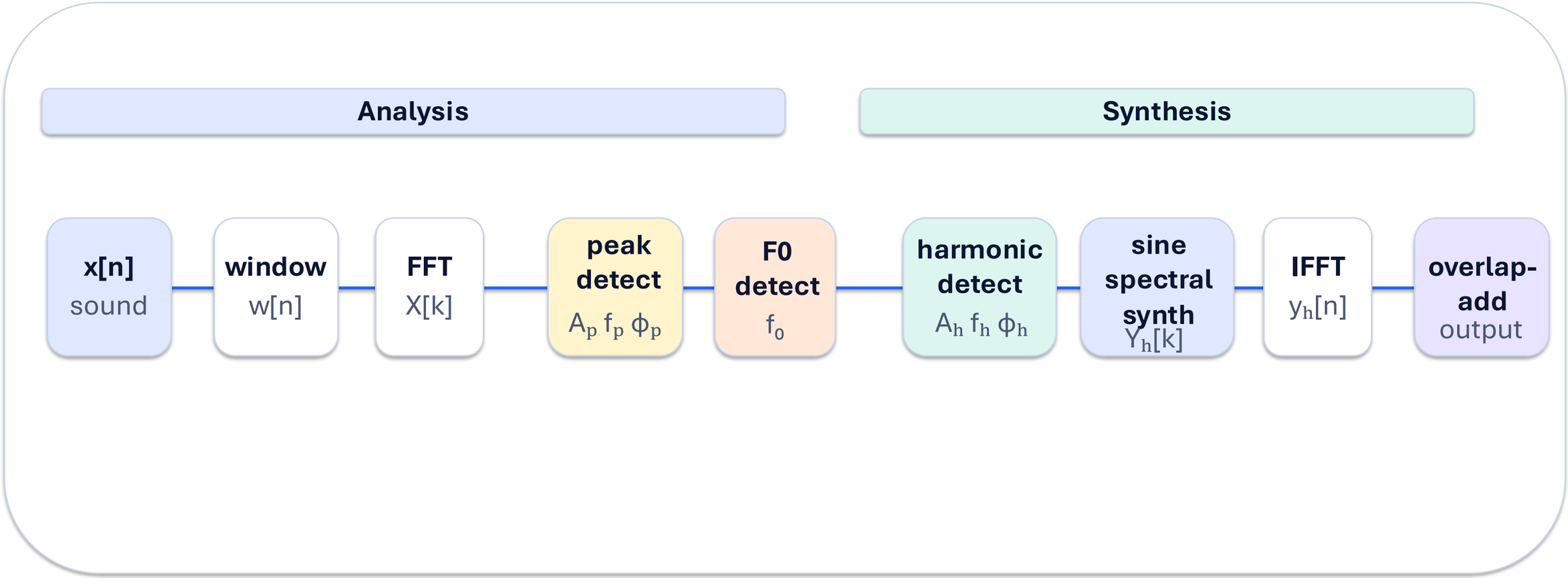
Common mistakes

- assuming the largest peak is always F0
- accepting every spectral peak as harmonic
- using F0 without checking peak deviation
- forgetting that noisy or weak peaks can break tracks

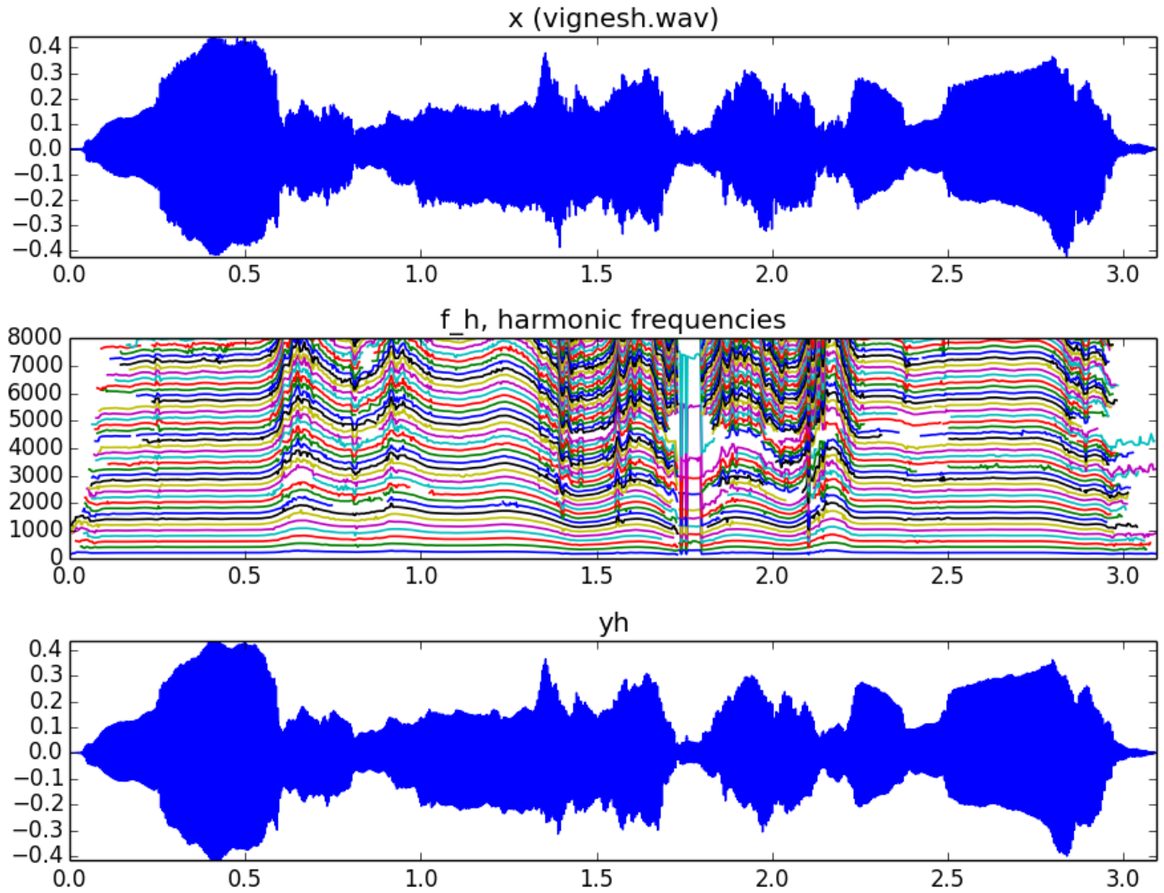
F0 explains the harmonic series; peak detection alone does not.

Harmonic model system

Analysis estimates harmonic parameters; synthesis turns those parameters back into a waveform



Harmonic analysis/synthesis of a voice sound



input waveform · harmonic frequencies · resynthesized waveform

Interpretation

- compare input and output waveforms
- middle plot shows harmonic frequencies
- gaps or wrong tracks become audible
- listening confirms model adequacy

Example notebook: examples/harmonic-model-examples.ipynb

Fundamental frequency and pitch

We use F0 and pitch together in class, but they are not exactly the same concept

F0

- signal-processing quantity
- frequency of the fundamental component or periodicity
- measured in Hz
- estimated from waveform or spectrum

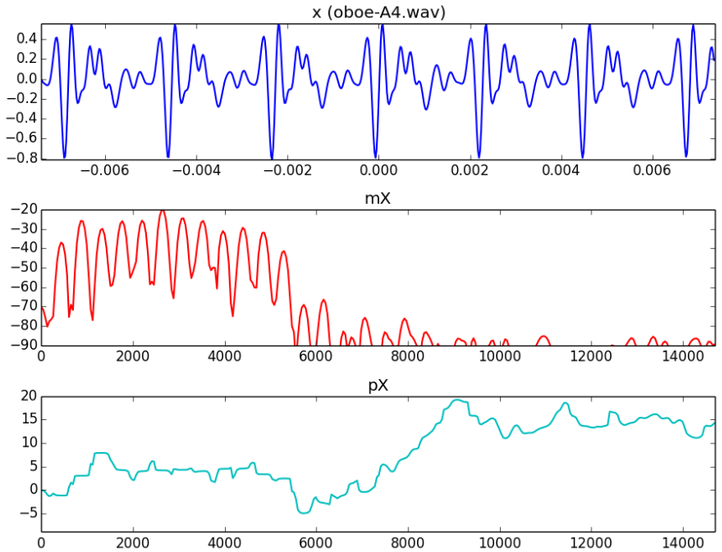
Pitch

- perceptual attribute
- what a listener hears as “high” or “low”
- related to F0, but not identical
- can be affected by timbre and context

Sound examples for F0 detection

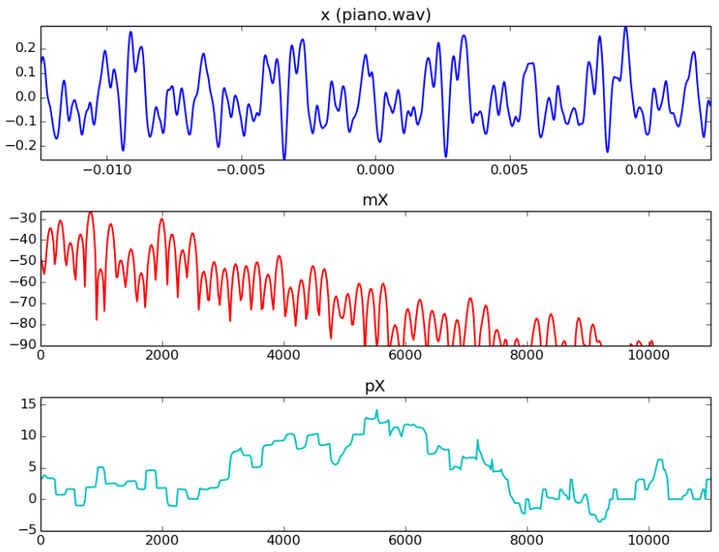
The same idea is much easier or harder depending on the sound and the mixture

oboe-A4.wav



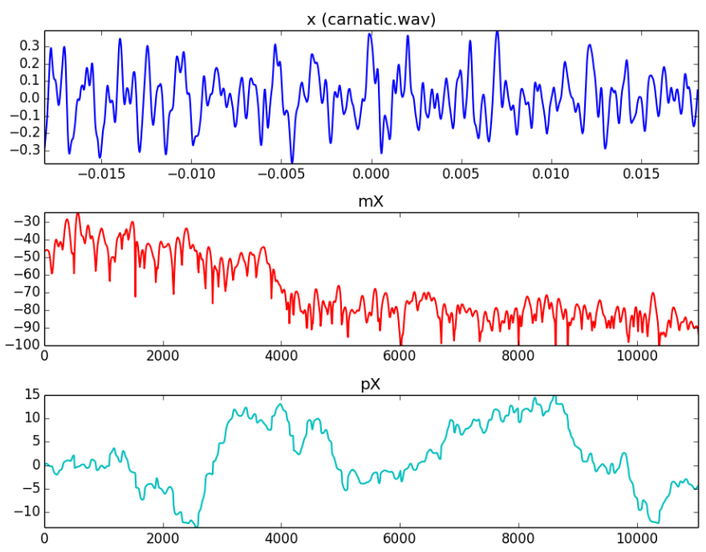
clear harmonic structure

piano.wav



time-domain detection is less clean

carnatic.wav



polyphonic mixture with a prominent voice

F0 detection with autocorrelation

In the time domain, F0 is estimated from the repeating period of the waveform

Autocorrelation idea

$$r_x[\ell] = \Sigma_n\, x[n]\, x[n-\ell] \quad \ell = 0, 1, \ldots, N-1$$

- compare the signal with delayed copies of itself
- large values indicate repeated structure
- the lag of repetition gives the period
- F0 is the inverse of the period in seconds

Where it works best

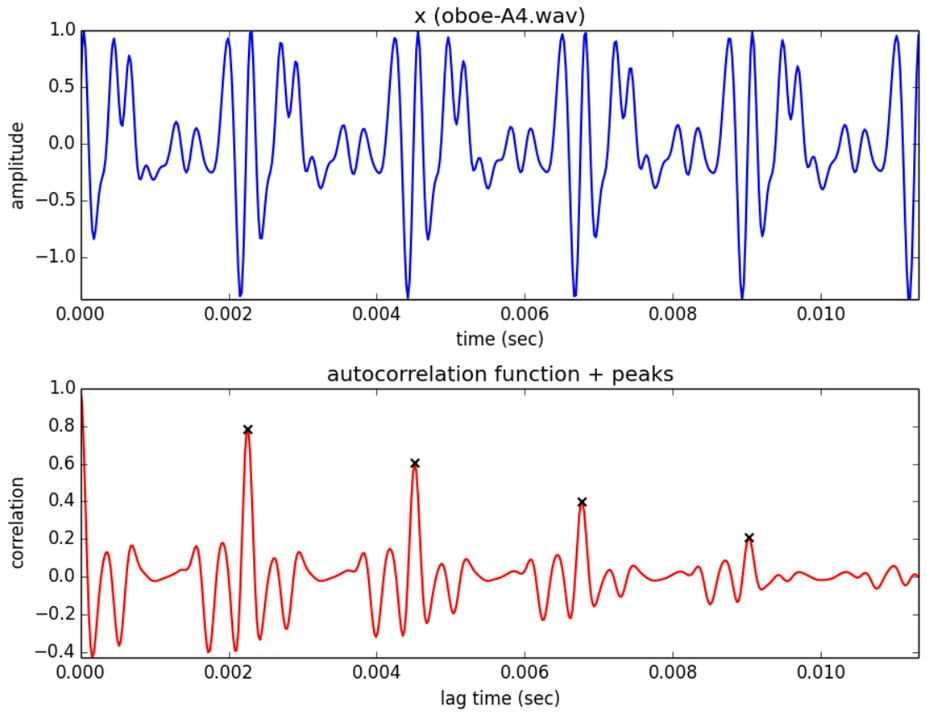
- monophonic sounds
- fairly stable periodicity
- clear cycles in the waveform
- less reliable with polyphony, noise, or strong transients

Autocorrelation estimates a period, not a spectral peak.

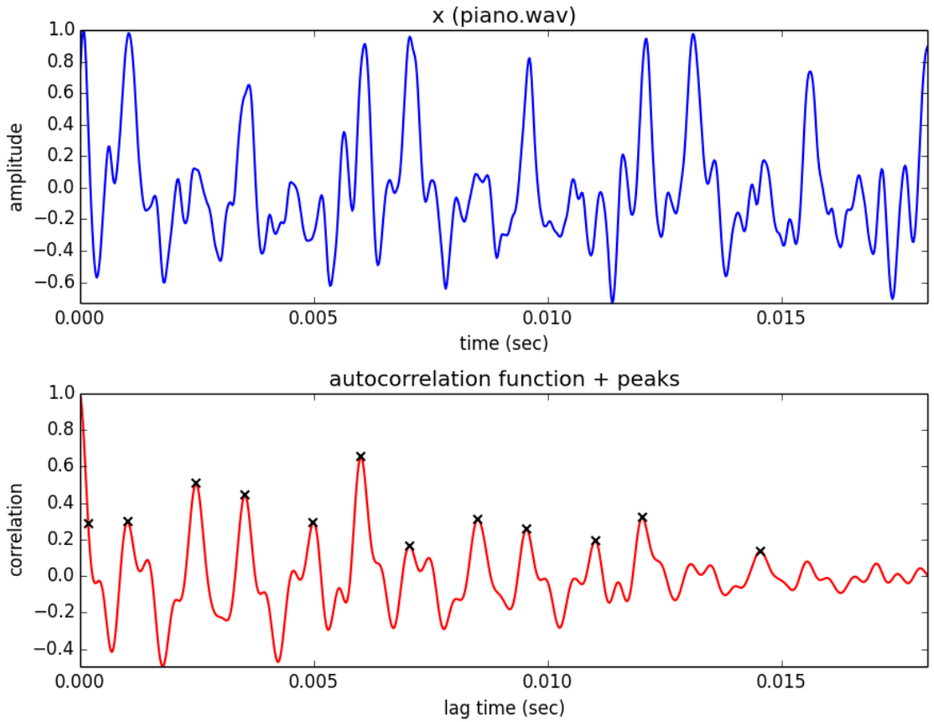
Autocorrelation examples: oboe vs piano

The same time-domain method can look clean for one instrument and ambiguous for another

Oboe: clearer periodicity



Piano: more ambiguous



YIN algorithm: difference function

YIN uses a related idea but looks for minima in a difference function

Difference function

$$d(\tau) = \Sigma_j \, (x_j - x_{j+}\tau)^2 \; \tau = 0, 1, \ldots, N-1$$

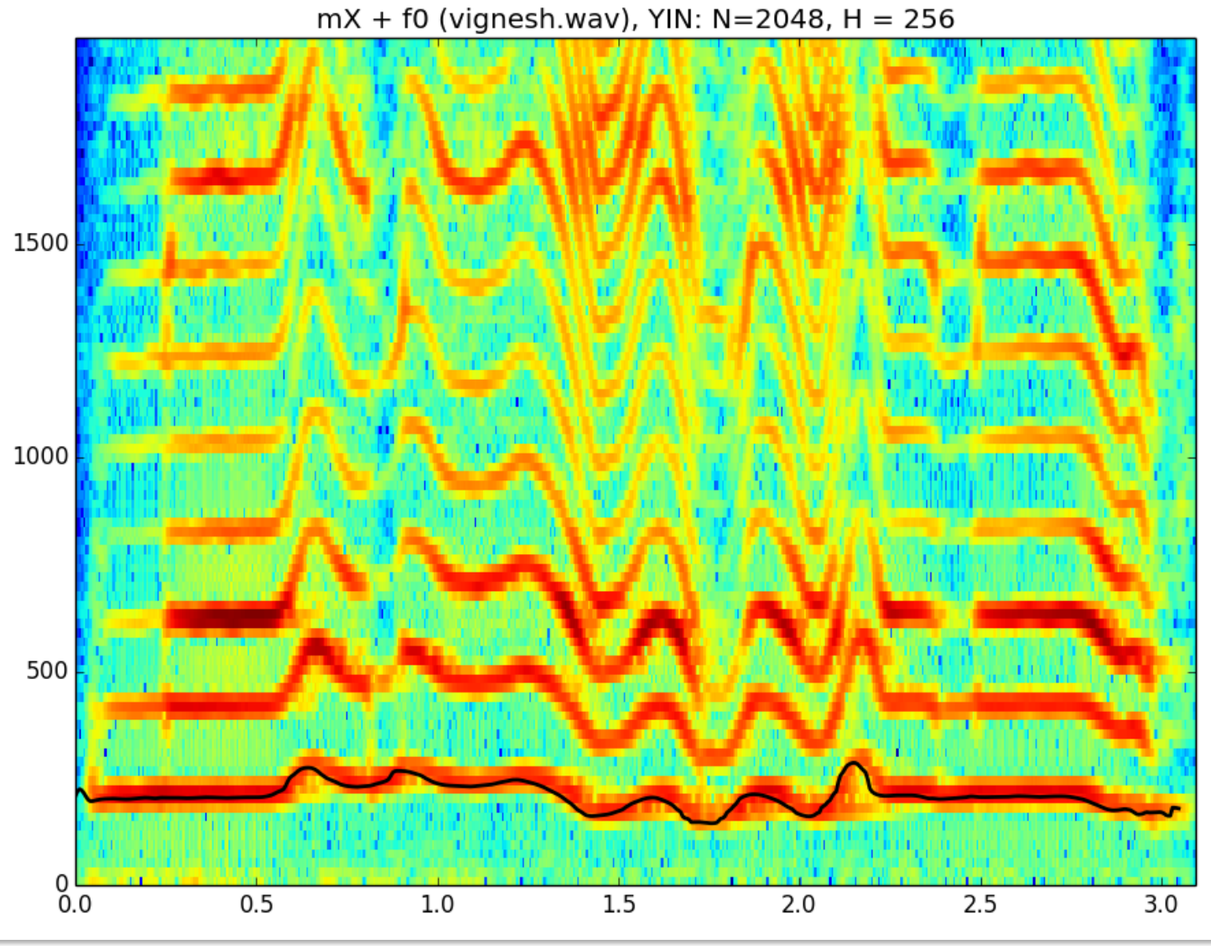
- small values mean the signal matches its delayed version
- minima are candidate periods
- the method is designed for monophonic F0 tracking
- still needs post-processing to avoid jumps

Conceptual contrast

- Autocorrelation: look for strong similarity peaks.
- YIN: look for low difference values.
- Both work directly in time domain.
- Both become difficult in mixtures.

Example notebook: examples/f0-detection.ipynb

YIN output over a vocal spectrogram

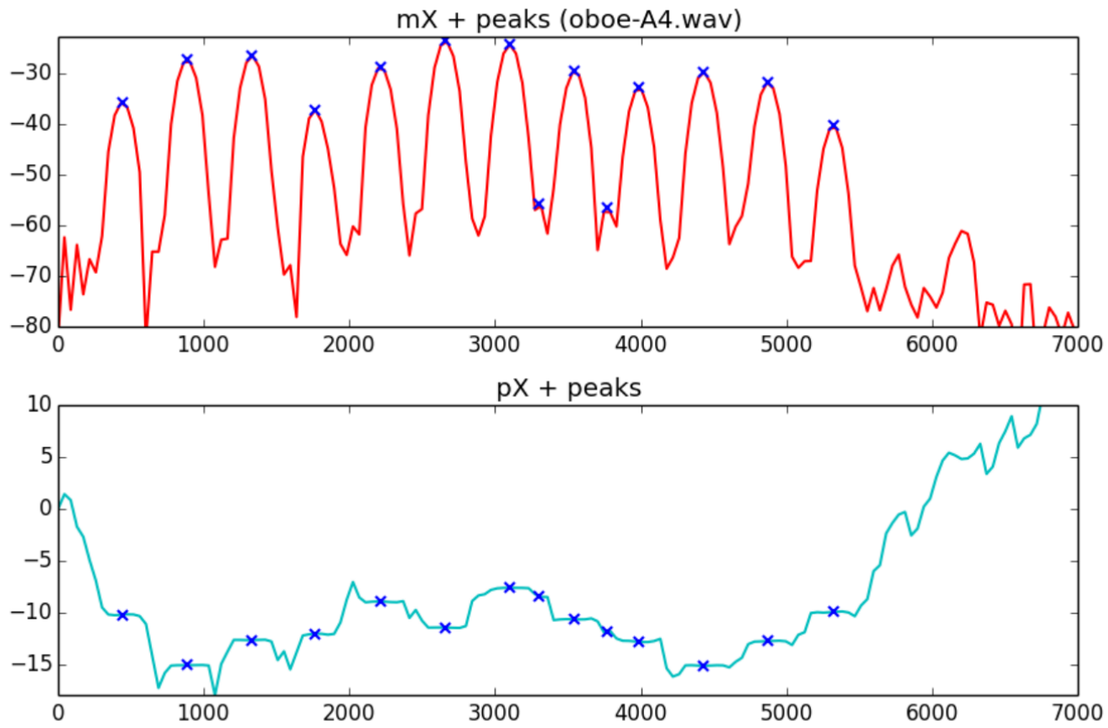


What to check

- does F0 follow the voice contour?
- are there octave jumps?
- are unvoiced/noisy regions ignored?
- does the curve align with harmonic ridges?

F0 detection from spectral peaks

In the spectrum, F0 is the common spacing that best explains the harmonic peaks



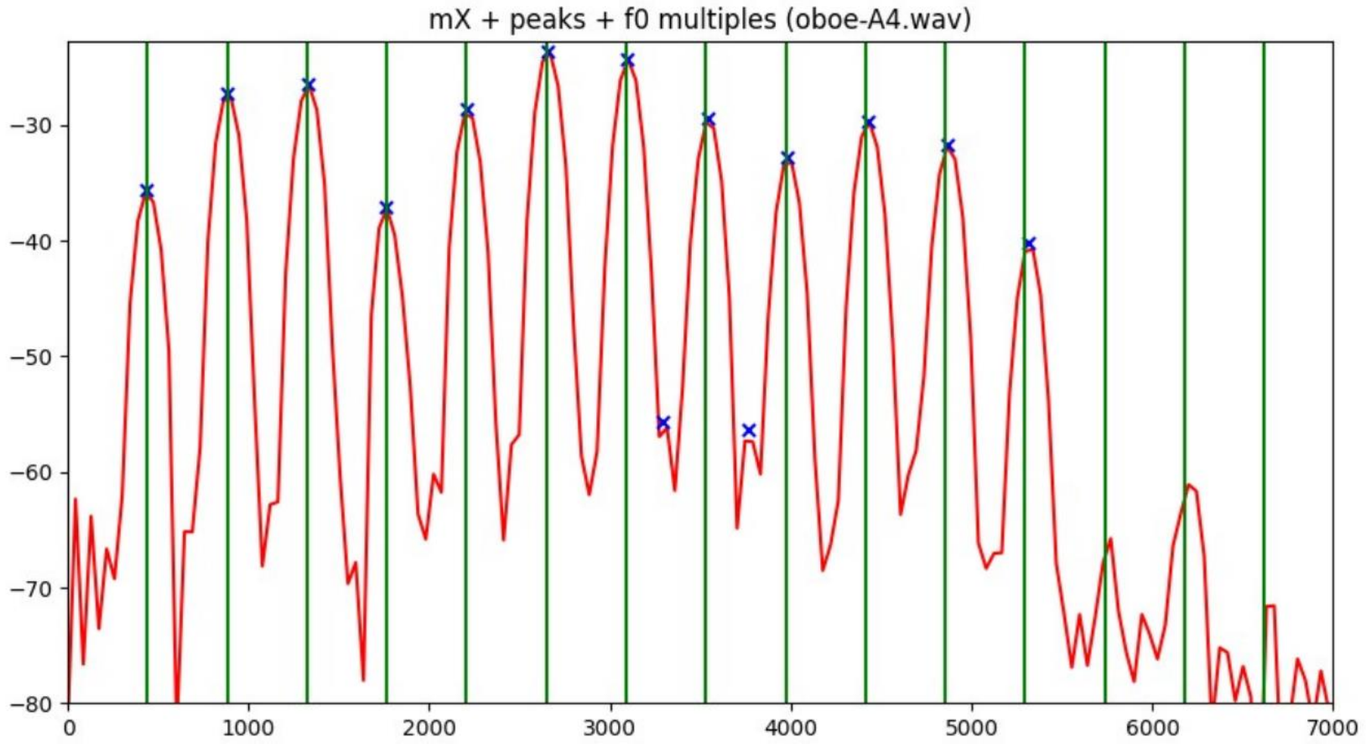
Core question

- Which harmonic series best explains the measured peaks?
- The lowest strong peak is not always F0, and F0 can be missing or weak.

F0 detection becomes a pattern-matching problem.

F0 as a common divisor in the spectrum

A harmonic F0 predicts peaks at F0, 2F0, 3F0, ...



Interpretation

- predicted harmonic locations form a regular grid
- measured peaks should fall near that grid
- extra peaks can be noise or another source
- missing harmonics do not necessarily invalidate the F0

Two-way mismatch as pattern matching

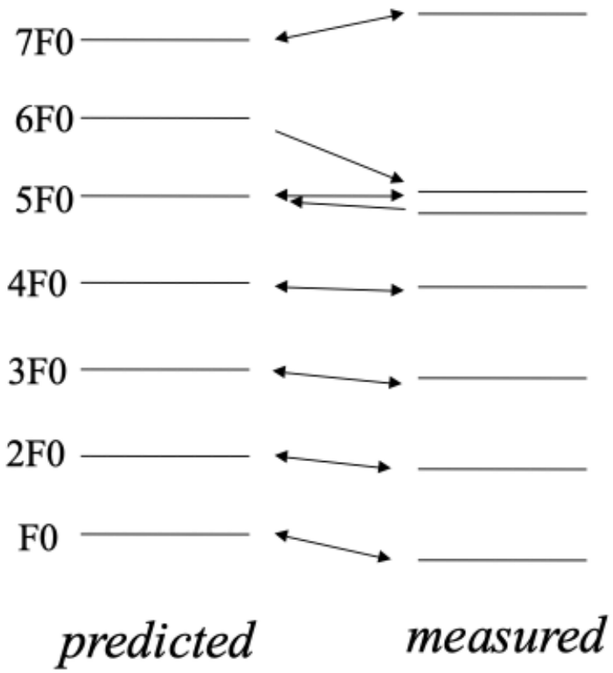
Compare predicted harmonic peaks with measured spectral peaks in both directions

$$\begin{aligned}\mathrm{Err}_{p\to m} &= \sum_{n=1}^{N} E_{\omega}\left(\Delta f_n, f_n, a_n, A_{\max}\right) \\ &= \sum_{n=1}^{N} \Delta f_n \cdot \left(f_n\right)^{-p} \\ &\quad + \left(\frac{a_n}{A_{\max}}\right) \times \left[q\,\Delta f_n \cdot \left(f_n\right)^{-p} - r\right]\end{aligned}$$

Δf_n : diff. between predicted and the closest measured peaks

f_n, a_n : frequency and magnitude of predicted peaks

$A_{\max}$: maximum peak magnitude



Two directions

1 · predicted → measured
Are predicted harmonics close to actual peaks?

2 · measured → predicted
Are measured peaks explained by the predicted harmonic series?

TWM error calculation

The algorithm scores candidate F0 values and selects the one with the lowest total error

Simplified total error

$$\mathrm{Err_total} = \mathrm{Err_p{\to}m} / N + \rho \cdot \mathrm{Err_m{\to}p} / K$$

Err_p→m: predicted-to-measured mismatch
Err_m→p: measured-to-predicted mismatch
N: number of predicted harmonics
K: number of measured peaks
ρ: weighting factor

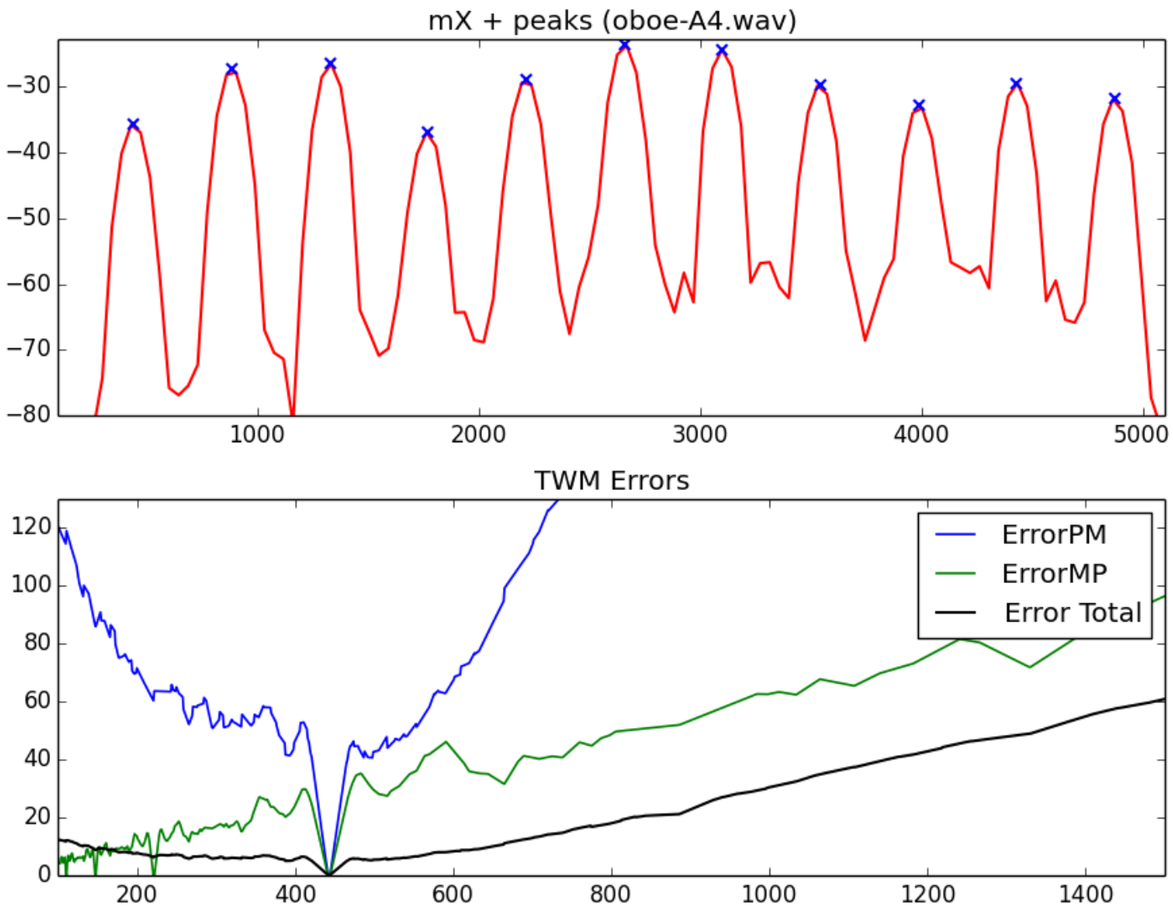
Example candidates

F0	Err p→m	Err m→p	Err total
50 Hz	122.58	−3.0	7.49
100 Hz	32.0	−3.0	3.83
200 Hz	10.0	30.66	4.2

For this example, 100 Hz has the lowest total error.

TWM error curves for the oboe sound

The selected F0 corresponds to the candidate with the minimum total mismatch

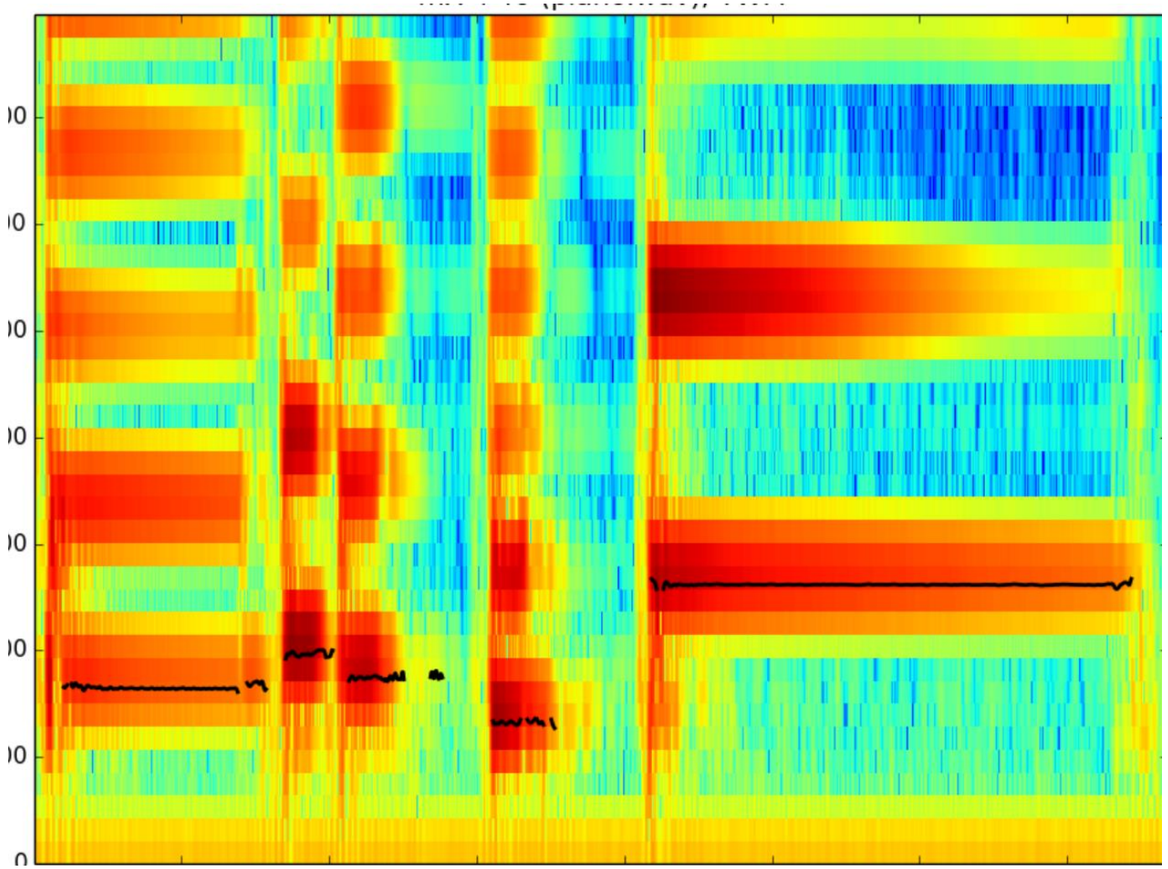


Reading the plot

- candidate F0 values are evaluated
- separate curves show mismatch components
- the minimum total error gives the estimate
- wrong octaves can create local minima

F0 trajectory for a piano phrase

A frame-level F0 estimate becomes a time-varying pitch contour

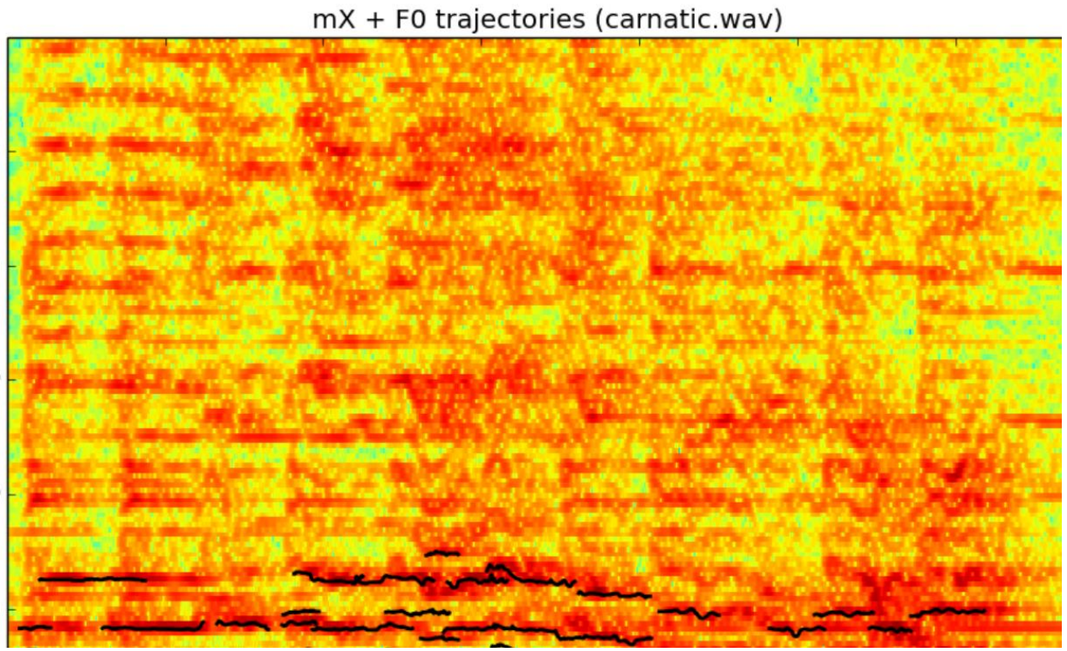


What can fail?

- octave errors
- noisy or missing estimates
- attacks with unstable spectra
- mixtures where several F0 values compete

F0 in polyphonic signals

In mixtures, several harmonic series can be present at the same time

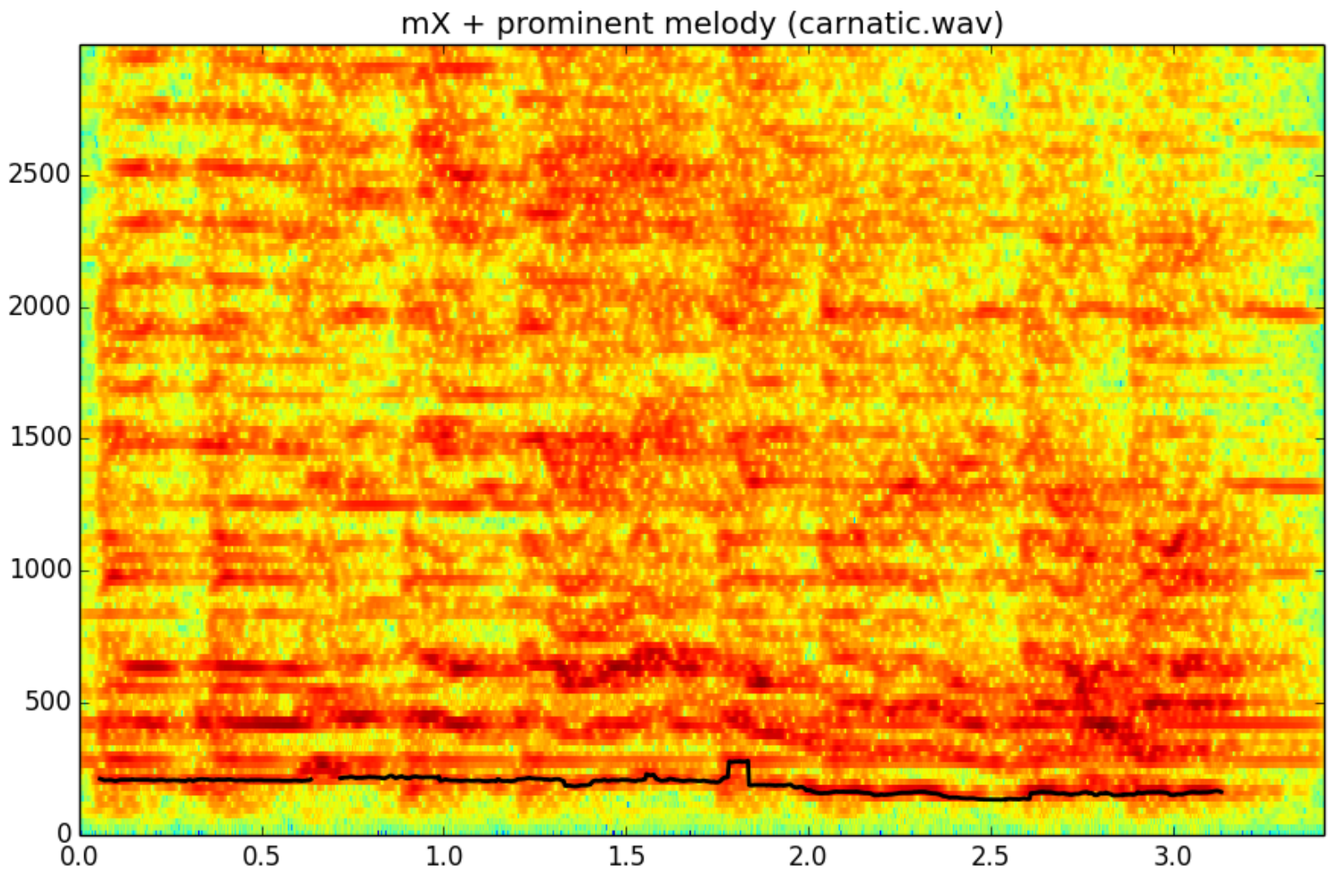


Why it is hard

- many sources may overlap in frequency
- some sources are inharmonic or percussive
- harmonics from different F0s can coincide
- a prominent melody may mask weaker instruments
- errors in F0 detection affect the whole harmonic model

Prominent pitch in polyphonic signals

A practical algorithm may estimate the dominant melodic pitch instead of all simultaneous F0s



Interpretation

- the curve follows the most salient pitch contour
- it is not a full transcription of all sources
- it is useful for melody extraction
- it illustrates why F0 detection is still an active problem

harmonicModelAnal(): waveform → harmonic tracks

Frame-wise DFT and spectral peaks, TWM F0 estimation, and F0-constrained peak selection

1 Window + DFT

Pad the signal, normalize w, then analyze overlapping frames.

mX, pX = dftAnal(x1, w, N)

2 Spectral peaks

Detect peaks above t dB; interpolate locations, magnitudes and phases.

ipfreq = fs · iploc / N

3 F0 (TWM)

Estimate f0t from interpolated peaks with f0Twm().

Use minf0, maxf0, f0et and the stable previous F0.

4 Select harmonics

harmonicDetection() checks peaks near r·F0 or previous-frame harmonics.

Keep up to nH below Nyquist.

Track continuity and cleaning

Previous-frame harmonics guide selection; cleaningSineTracks() removes segments shorter than minSineDur.

Returns: one row per analysis frame

xhfreq (Hz) · xhmag (dB) · xhphase (rad); each array has nFrames × nH entries.

Example notebooks

1

examples/f0-detection.ipynb

Fundamental-frequency detection: candidate F0, error curves, and contours.

2

examples/harmonic-model-examples.ipynb

Harmonic analysis/synthesis: F0, harmonic selection, tracks, and resynthesis.

3

examples/harmonicModel_function.py / GUI

Compact function call: parameters, output sound, and diagnostic plots.

How to read the example outputs

Interpret each harmonic-model plot together with its F0, parameters, and sound

1 · F0

Does the estimated F0 explain the spectral peaks?

2 · Harmonics

Are the selected peaks close to integer multiples of F0?

3 · Tracks

Are the harmonic tracks stable enough to resynthesize the source?

4 · Resynthesis

Does the output preserve pitch, amplitude evolution, and timbral identity?

Exercise 6: Harmonic model

Sixth lab: estimate F0, select harmonics, synthesize the harmonic signal, and diagnose errors

What you will practice

- estimate F0 from spectral peaks
- select harmonics using an F0 constraint
- compare tracks, waveform, and sound

How to start

Open: exercises/E6-Harmonic-model.ipynb

Use: f0Detection() · harmonicModelAnal() · harmonicModelSynth()

1 F0 — 2 Peaks — 3 Tracks — 4 Synth — 5 Check

Exercise 6: change parameters deliberately

Each parameter affects what is detected, tracked, synthesized, and heard

parameter	meaning	what to observe
minf0 / maxf0	F0 search range	wrong range causes octave errors or no estimate
f0et	F0 error threshold	too strict rejects frames; too loose accepts wrong F0
harmDevSlope	harmonic deviation tolerance	larger tolerance keeps unstable harmonics but may add wrong peaks
t	peak threshold	controls weak/noisy peaks before harmonic selection
maxnHarm	harmonic limit	sets maximum number of harmonics used per frame
minSineDur	track cleaning	removes short tracks that may be spurious

What to remember from Topic 6

Harmonic modeling combines sinusoidal analysis with an F0 constraint

1 Harmonic model

sinusoidal frequencies are multiples of F0

2 Harmonic detection

select peaks that fit the harmonic series

3 F0 detection

estimate the fundamental from time or frequency evidence

4 TWM

match predicted harmonics and measured peaks

5 Polyphony

mixtures make F0 detection and harmonic selection harder

6 Validation

inspect F0, tracks, spectrum, waveform, and listening

References and credits

- sms-tools models: https://github.com/MTG/sms-tools
- sms-tools-materials repository: notebooks, lecture plots, lab materials, and sounds
  - examples/f0-detection.ipynb for autocorrelation, YIN, and TWM-based F0 estimation
  - examples/harmonic-model-examples.ipynb for harmonic analysis, tracking, and synthesis
  - exercises/E6-Harmonic-model.ipynb for the practical lab
- de Cheveigné and Kawahara (2002), “YIN, a fundamental frequency estimator for speech and music.”
- Maher and Beauchamp (1994), “Fundamental frequency estimation using a Two-Way Mismatch procedure.”
- Salamon and Gómez (2012), “Melody extraction from polyphonic music signals using pitch contour characteristics.”
- Slides released under CC Attribution-Non Commercial-Share Alike license and code under Affero GPL license.